Quiz 3 (Take home)

1.) Liquid → keep in concentrations

$$\frac{dN_A}{dt} = F_{A0} - F_A + r_A V$$

$$F_{A0} - F_A = v_0 (C_{A0} - C_A)$$

$$\frac{dC_A}{dt} = r_A + \frac{v_0 (C_{A0} - C_A)}{V}$$

$$r_A = -r_B = -k C_A$$

$$\frac{dN_B}{dt} = \cancelto{0}{F_{B0}} - \cancelto{0}{F_B} + r_B V$$

$$\frac{d(C_B V)}{dt} = V \frac{dC_B}{dt} + v_0 C_B$$

$$\frac{dC_B}{dt} = r_B - \frac{v_0 C_B}{V}$$

s Settings Help

Untitled.ipynb | P8-5.ipynb | CHE 321 Quiz 3 Code.ipy | P8-11-PFR.ipynb

Code

[1]:
```
import numpy as np
from scipy.integrate import odeint
from scipy.optimize import fsolve
import matplotlib.pyplot as plt
from scipy.interpolate import interp1d
import matplotlib
```

[15]:
```
k = .4 # [1/min]
CA0 = .05 #mol/dm3 molar conc of inlet
v0i = 2 #dm^3/min initial volumetric flow rt
alpha = 1 #min-1
V0 = 2 #dm^3 intial volume of reactor

def ODEfun(C, t, k, CA0, v0i, alpha, V0):

    CA = C[0]
    CB = C[1]

    rA = - k * CA
    rB =  -rA

    exp = np.exp(-alpha*t)
    v0 = v0i * exp
    V = V0 + v0 * t

    dCAdt = rA + v0 * (CA0 - CA) / V
    dCBdt = rB - v0 * CB / V

    return np.array([dCAdt, dCBdt])
```

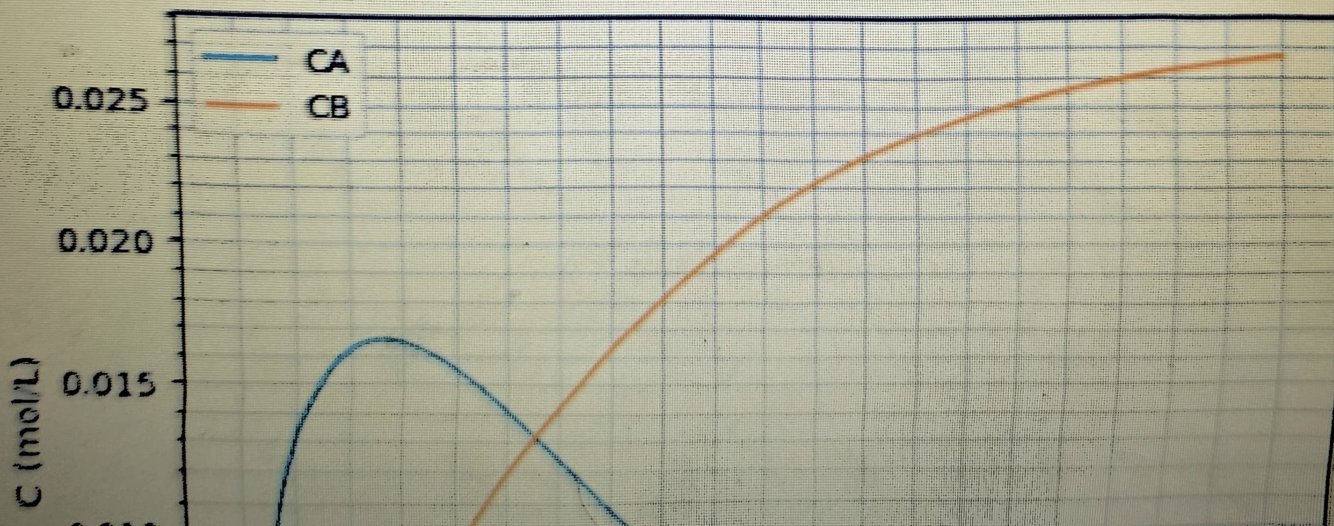
[16]:
```
t = np.linspace(0, 10, 1000)
```

nv:base] * | Idle

Search

[16]:
```python
t = np.linspace(0, 10, 1000)
CAi = 0
CBi = 0
inivals = np.array([CAi, CBi])
solution = odeint(ODEfun, inivals, t, (k, CA0, v0i, alpha, V0))
CA = solution[:, 0]
CB = solution[:, 1]
```

[18]:
```python
fig, ax = plt.subplots()
ax.plot(t, CA, label='CA')
ax.plot(t, CB, label='CB')
ax.set_xlabel('t (min)')
ax.set_ylabel('C (mol/L)')
ax.legend()
ax.grid()
ax.grid(which = "minor")
ax.minorticks_on()
```



```
ax.set_xlabel('t (min)')
ax.set_ylabel('C (mol/L)')
ax.legend()
ax.grid()
ax.grid(which = "minor")
ax.minorticks_on()
```